Sparse Learning and Endogenous Pockets of Predictability

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Abstract

We develop an asset-pricing model in which investors search for return predictability in high-dimensional data using a LASSO-based learning rule that mirrors empirical practices. Although returns are unpredictable under full-information rational expectations, investors test this hypothesis by regressing returns on a vast array of signals. We show that this behavior induces a nonlinear feedback from beliefs to prices that prevents convergence to rational expectations and instead generates transient, sparse episodes of predictability. The key implication is that the search for predictability itself endogenously generates short-lived windows of predictability. We characterize these dynamics analytically and document their empirical relevance in the cross-section of U.S. stock returns.

Keywords: Asset pricing, learning, LASSO, return predictability, high-dimensional forecasting.
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1 Introduction

The volume and granularity of data generated worldwide have expanded at an unprecedented pace. In 2024 alone, global data creation reached approximately 402 million terabytes per day, up from about 43 million a decade earlier (Wright, 2025). Financial markets have mirrored this expansion. Whereas investors once relied on a limited set of economic and accounting indicators, they now draw on a broad and rapidly growing array of signals, including textual data, satellite imagery, and web-based activity. For most of these variables, however, economic theory offers little guidance on their relevance for forecasting stock returns. Despite this lack of theoretical discipline, demand for alternative data remains strong, as investors increasingly employ machine learning techniques to search for predictive patterns within high-dimensional datasets (Invesco, 2024). These methods work without prior restrictions on which variables should matter, allowing the data to guide model selection.

This behavior is not well captured by existing asset-pricing models with learning. In such frameworks, agents are assumed to learn using some perceived law of motion specified by the modeler. As a result, learning is confined to tracking a small number of parameters, which are updated recursively using variants of least squares or constant-gain algorithms. While analytically convenient, these approaches provide a poor description of modern investment behavior, where investors routinely consider thousands of potential signals without committing to a specific structural model.

We develop a model that embeds statistical learning into an otherwise standard asset-pricing environment. Agents in our model suspect that returns may be conditionally predictable from a high-dimensional set of observable predictors. To exploit this perceived opportunity, they employ a learning rule that is based on the least absolute shrinkage and selection operator (LASSO) (Tibshirani, 1996). Agents first estimate a perceived law of motion (PLM) using least squares and then reduce the dimensionality of their model by shrinking all coefficients to zero that are below a certain threshold.

Our main result is that the act of searching for predictability among many candidate predictors can itself generate short-lived pockets of return predictability. In equilibrium, these episodes are sparse and short-lived, in line with recent evidence on episodic short-horizon predictability (Chinco, Clark-Joseph and Ye, 2019; Farmer, Schmidt and Timmermann, 2023). The mechanism follows a simple life cycle. In finite samples, the learning rule occasionally selects predictors with estimated coefficients large enough to survive the shrinkage. When investors trade on this perceived predictability, their demand feeds back into prices, transforming the perceived signal into genuine but temporary return predictability. As new data arrive and older observations are discounted, the selected coefficients revert toward zero, the signal dissipates, and prices return to behavior consistent with unconditional unpredictability.

We reach this result, by formalizing the mechanism in a Lucas-type exchange economy with a single risky asset. Under rational expectations, the price–dividend ratio is constant and returns are unpredictable. We follow Adam, Marcet and Nicolini (2016) and depart from rational expectations by assuming that agents know the dividend and consumption process but not the process for returns. Instead, they suspect that returns are conditionally predictable from a large set of observable signals. Each period, investors estimate a PLM in which next periods return depends linearly on a set of predictors. As the number of candidate predictors exceeds the sample size by a large margin, standard linear estimation techniques are infeasible and the agent estimates the PLM using a soft-thresholding rule similiar to the LASSO. Beliefs then feed into equilibrium prices through the actual law of motion (ALM) governing returns. In contrast to the linear PLM, the ALM is nonlinear and thus misspecified. In this case, learning is unlikely to converge to the true law of motion. Instead, the economy converges to a cross-moment consistent equilibrium (CMCE) in which agents estimated forecasting moments coincide with those generated by the equilibrium itself. Occasional variable-selection events can then trigger predictable return episodes that persist only until the corresponding coefficients shrink back toward zero.

We test the models predictions using a large sample of NYSE-listed stocks and a set of 180 news attention series constructed in Bybee et al. (2024). Our estimations confirm several key implications of the model. First, we document that LASSO uncovers episodic return predictability concentrated in a small and shifting set of predictors, consistent with Chinco, Clark-Joseph and Ye (2019). Second, we show that the frequency of these episodes is related to the volatility of the return process. Third, we find that our model can explain a significant fraction of the cross-sectional variation in stock returns.

The rest of the paper is structured as follows. Section 2 discusses related literature. Section 3 presents the model environment. Section 4 analyzes the asset-pricing equilibrium under recursive least squares. Section 5 analyzes the asset-pricing equilibrium under weighted least squares. Section 6 introduces LASSO learning and derives our main results. Section 7 estimates the model on a sample of daily stock returns. Section 8 concludes.

2 Related Literature

"This paper relates to x strands of literature…"

Learning is often introduced in asset pricing models to account for empirical features such as excess volatility or return predictability that are difficult to reconcile with fully informed agents. The central premise of this literature is that investors have incomplete knowledge of the data generating process governing market outcomes and therefore infer it from observed data. Our model relates to this large literature on asset pricing with learning.

Early contributions assume fully rational agents who estimate unknown parameters of a correctly specified model using least squares (Marcet and Sargent, 1989). In this setting, learning converges to the rational expectations equilibrium as agents accumulate information over time. Even in such low-dimensional environments, learning can generate excess volatility and return predictability (Timmermann, 1993, 1996). Convergence to the rational expectations equilibrium is no longer guaranteed when agents are boundedly rational, as finite-memory and finite-sample learning can lead beliefs to fail to converge (Honkapohja and Mitra, 2003). The implications of learning become richer when agents face model uncertainty in addition to parameter uncertainty. Barberis, Shleifer and Vishny (1998) study investors who switch between competing forecasting rules, while Hong, Stein and Yu (2007) and Branch and Evans (2010) analyze settings in which agents estimate univariate perceived laws of motion although the true data generating process is multivariate.

More recently, Adam and Marcet (2011) introduce the concept of *internal rationality*, whereby agents choose actions optimally given their subjective beliefs, even when those beliefs are misspecified. Building on this insight, Adam, Marcet and Nicolini (2016) embed internally rational agents into a standard asset-pricing framework in which investors learn the mapping from fundamentals to prices, generating belief–price feedback that helps rationalize key asset-pricing moments. We adopt this framework and its central feedback mechanism, but replace their general learning rule with a microfounded LASSO-based learning rule. This modification allows us to study how high-dimensional statistical learning and endogenous model selection shape equilibrium price dynamics.

More broadly, our paper also relates to research on how the growth of data affects asset prices. Begenau, Farboodi and Veldkamp (2018) show that greater data-processing capacity improves investor forecasts and alters equilibrium prices and the cost of capital, while Farboodi and Veldkamp (2020) study investors who optimally choose whether to acquire information about fundamentals or about the behavior of other market participants.

On the empirical side, our work relates to the extensive literature on time-varying stock return predictability. A wide range of variables, including consumption growth and valuation ratios, forecast returns in some periods but not in others (Ang and Bekaert, 2007; Goyal and Welch, 2003; Lettau and Ludvigson, 2001). A complementary line of research emphasizes sparsity. Among the many predictors proposed in the literature, only a small and time-varying subset is informative at any given point in time (Kozak, Nagel and Santosh, 2020; Freyberger, Neuhierl and Weber, 2020). Consistent with this view, Mclean and Pontiff (2016) document that many return anomalies attenuate following publication, suggesting that investors actively search for, learn about, and trade on such signals. We add to this empirical literature by providing a theoretical mechanism for the emergence and disappearance of sparse return predictability.

Finally, our paper contributes to the growing literature on the use of machine learning in finance. This literature has largely focused on the empirical gains from machine-learning methods for return forecasting and portfolio choice (Feng, Giglio and Xiu, 2020). Gu, Kelly and Xiu (2020) show that machine-learning forecasts outperform traditional linear models by exploiting sparsity and nonlinear interactions among predictors. Most closely related to our learning mechanism, Chinco, Clark-Joseph and Ye (2019) document that LASSO uncovers short-lived pockets of high-frequency predictability concentrated in a small and shifting set of predictors. Our contribution is to shift the focus from forecasting performance to the equilibrium consequences of investors using machine learning techniques in their trading decisions.

3 The model

We study a standard Lucas (1978) asset pricing model in the flavour of Adam, Marcet and Nicolini (2016). We consider an economy populated by a continuum of agents of mass one who trade a single risky asset. The asset pays a stochastic dividend D_t at each discrete time period $t = 0, 1, 2, \ldots$. The dividend and consumption processes are driven by multivariate normally distributed shocks

$$\begin{pmatrix} \eta_t^d \\ \eta_t^c \end{pmatrix} = \begin{pmatrix} \log \varepsilon_t^d \\ \log \varepsilon_t^c \end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix} -\sigma_d^2/2 \\ -\sigma_c^2/2 \end{pmatrix}, \begin{pmatrix} \sigma_d^2 & \rho\sigma_d\sigma_c \\ \rho\sigma_d\sigma_c & \sigma_c^2 \end{pmatrix} \right), \tag{1}$$

The dividend process follows

$$\frac{D_t}{D_{t-1}} = a\varepsilon_t^d, \tag{2}$$

where $a \geq 1$ is a constant growth factor, which implies that

$$\mathbb{E}_t\left[D_{t+1}\right] = aD_t. \tag{3}$$

Similarly for the aggregate consumption process we postulate that

$$\frac{C_t}{C_{t-1}} = a\varepsilon_t^c. \tag{4}$$

The equilibrium price of the asset is determined by the standard no-arbitrage condition which requires that the price at time t equals the discounted expected value of next period's price plus dividend, that is

$$P_t = \delta\mathbb{E}_t\left[\left(\frac{C_{t+1}}{C_t} \right)^{-\ell} P_{t+1} + D_{t+1} \right], \tag{5}$$

where δ is the discount factor, ℓ is the risk aversion parameter[1], and $\mathbb{E}_t[\cdot]$ is the expectation operator conditional on information available at time t. Under full information rational expectations (FIRE), the equilibrium price can be solved as

$$P_t = \frac{\delta a^{(1-\ell)}\rho_e}{1 - a^{(1-\ell)}\rho_e} D_t, \tag{6}$$

where

$$\rho_e = \mathbb{E}_t\left[\left(\frac{C_{t+1}}{C_t} \right)^{-\ell} \frac{D_{t+1}}{D_t} \right] = e^{\ell(1+\ell)\sigma_c^2/2} e^{-\ell\rho\sigma_c\sigma_d}. \tag{7}$$

[1] The consumption based asset pricing literature normally used γ, but we will use ℓ to avoid confusion with the learning gain parameter

This implies a constant price-dividend ratio and an ex-dividend gross return which is given by

$$R_{t+1} = \frac{P_{t+1}}{P_t} = \frac{D_{t+1}}{D_t} = a\varepsilon^d_{t+1}. \tag{8}$$

Taking logs we obtain log-returns

$$r_{t+1} = \log(R_{t+1}) = \log(a) + \eta^d_{t+1} \sim \mathcal{N}(\log(a) - \sigma^2_d/2, \sigma^2_d). \tag{9}$$

Without loss of generality in what follows we set $\log(a) = \sigma^2_d/2$. This implies that under FIRE log-returns have mean zero and there is no time-t information that can help forecasting returns.

Instead of assuming that agents know the true data generating process, we follow the adaptive learning literature and assume that agents are boundedly rational and form their expectations using econometric techniques (Evans and Honkapohja, 2001). Equation (5) shows that agents have to forecast two quantities, the risk-adjusted gross return and gross dividend growth. We follow Adam, Marcet and Nicolini (2016) in making the next assumption.

Assumption 1 *Agents know the process for dividends and consumption and are boundedly rational only with respect to (gross) returns.*

In this way we can isolate the effects of learning about returns on asset prices.

3.1 Univariate PLM and ALM

To fix ideas we start by considering the univariate case in which agents try to predict log-returns using a single predictor x_t. Specifically we assume that agents have a linear and stationary PLM for the asset return

$$r_{t+1} = x_t\beta + u_{t+1}, \tag{10}$$

with $x_t \sim \mathcal{N}(0, \sigma^2_x)$ being a signal available for the agent at time t which they suspect might have predictive power on log-returns. Then the agents' PLM about log-returns implies the following PLM for gross returns

$$R_{t+1} = e^{x_t\beta + u_{t+1}}. \tag{11}$$

This implies that gross expected[2] returns follow a log-normal distribution and their conditional forecast is

$$\tilde{\mathbb{E}}_t(R_{t+1}) = \tilde{\mathbb{E}}_t\left(\frac{P_{t+1}}{P_t}\right) = e^{x_t\beta + \sigma^2_u/2} = \phi e^{x_t\beta}, \tag{12}$$

by letting $\phi = e^{\sigma^2_u/2}$.

[2]Throughout the paper we use $\tilde{\mathbb{E}}_t[\cdot]$ to denote expectations taken under the agents' PLM, while $\mathbb{E}_t[\cdot]$ denotes rational expectations.

Assumption 2 *Within their perceived law of motion, agents believe that the return process and consumption growth are independent.*

Assumption 2 allows us to derive risk-adjusted expected gross returns as

$$\tilde{\mathbb{E}}_t\left[\left(\frac{C_{t+1}}{C_t}\right)^{-\ell}\frac{P_{t+1}}{P_t}\right]=\tilde{\mathbb{E}}_t\left[\left(\frac{C_{t+1}}{C_t}\right)^{-\ell}\right]\tilde{\mathbb{E}}_t\left(\frac{P_{t+1}}{P_t}\right)=a^{-\ell}\phi e^{x_t\beta}. \tag{13}$$

Plugging this into (5) yields

$$P_t=\frac{\delta a^{(1-\ell)}\rho_e}{1-\delta a^{-\ell}\phi e^{x_t\beta}}D_t. \tag{14}$$

Now divide by P_{t-1} to obtain actual gross returns

$$R_t=\frac{P_t}{P_{t-1}}=\frac{1-\delta a^{-\ell}\phi e^{x_{t-1}\beta}}{1-\delta a^{-\ell}\phi e^{x_t\beta}}\frac{D_t}{D_{t-1}}, \tag{15}$$

and taking logs and moving one step forward we obtain the ALM

$$r_{t+1}=\eta_{t+1}+\log(1-\kappa e^{x_t\beta})-\log(1-\kappa e^{x_{t+1}\beta}), \tag{16}$$

where $\kappa:=\delta a^{-\ell}\phi$ and $\eta_{t+1}=\log(a)+\eta^d_{t+1}$.

Expression (16) is only well-defined when $1-\kappa e^{x_t\beta}>0$ and $1-\kappa e^{x_{t+1}\beta}>0$ hold with probability one, which imposes state-dependent restrictions on (x_t,β). To avoid these domain issues and guarantee global existence of a stochastic equilibrium, we adopt a bounded specification for the conditional mean of returns. We modify the perceived conditional expectation of gross returns to

$$\tilde{\mathbb{E}}_t(R_{t+1})=\phi\exp(w\tanh(x_t\beta/w)), \tag{17}$$

where $w=-\log(\kappa)-\varepsilon$ for some $\varepsilon>0$. Since $\tanh(\cdot)\in(-1,1)$, we have $w\tanh(x_t\beta/w)\in(-w,w)$ which implies $e^{w\tanh(x_t\beta/w)}\in(e^{-w},e^w)$. By construction $w<-\log(\kappa)$, so $e^w<\kappa^{-1}$ and therefore $1-\kappa e^{w\tanh(x_t\beta/w)}>0$ for all x_t and β.

The corresponding risk-adjusted expectations and price-dividend ratio are defined as before, with $x_t\beta$ replaced by $w\tanh(x_t\beta/w)$. This yields the modified ALM

$$r_{t+1}=\eta_{t+1}+\log(1-\kappa e^{w\tanh(x_t\beta/w)})-\log(1-\kappa e^{w\tanh(x_{t+1}\beta/w)}). \tag{18}$$

For later use, we define the function

$$g_\beta(x):=\log(1-\kappa\, e^{w\tanh(x\beta/w)}), \tag{19}$$

which is globally well-defined and smooth in x and β. Given that $X_t\beta_t$ represents expected returns, which in practice move within a limited range rather than taking arbitrarily large values, we argue that the original ALM is well defined with high probability. The bounded specification (18), should therefore be interpreted as a globally valid extension that coincides with the baseline model in the empirically relevant region of the state space while eliminating corner cases where the logarithm is undefined. As noted above, agents do not know the value of β in their PLM (10) and have to estimate it from observed data. We therefore consider different learning mechaisms that agents can use to update their beliefs about β over time, sequentially building towards the LASSO learning mechanism that is the main focus of this paper.

4 Recursive Least Squares Learning

We first consider Recursive Least Squares (RLS) learning, which is a recursive implementation of standard Ordinary Least Squares (OLS). At each time t, agents update their estimate β_t by minimizing the sum of squared forecast errors:[3]

$$\beta_t = \left(\sum_{i=1}^{t-1} x_{i-1}^2\right)^{-1} \sum_{i=1}^{t-1} x_{i-1} r_i. \tag{20}$$

This estimator admits a recursive formulation. Define $M_{t-1} = \frac{1}{t-1}\sum_{i=1}^{t-1} x_{i-1}^2$. Standard algebra, which we defer to Appendix (A.1) yields the recursions

$$M_t = M_{t-1} + \frac{1}{t}\left(x_{t-1}^2 - M_{t-1}\right), \tag{21}$$

$$\beta_t = \beta_{t-1} + \left((t-1)M_{t-1}\right)^{-1}\left(x_{t-2}r_{t-1} - x_{t-2}^2\beta_{t-1}\right). \tag{22}$$

Under RLS learning, actual returns evolve according to

$$r_{t-1} = \eta_{t-1} + g_{\beta_{t-2}}(x_{t-2}) - g_{\beta_{t-1}}(x_{t-1}), \tag{23}$$

where $g_\beta(\cdot)$ is defined in (19).

A natural question is whether this learning process converges to the rational expectations equilibrium. Under FIRE, returns in our model are unpredictable and $\beta = 0$. However, convergence to the REE requires not only that agents' perceived relationship matches the actual relationship, but that the entire distributions implied by the PLM and ALM coincide. When the ALM is nonlinear and the PLM is linear, such convergence is generally not possible. The learning literature has instead focused on weaker notions of equilibrium that require consistency only on specific moments rather than on entire distributions (Hommes and Zhu, 2014). Following this approach, we define a Cross-Moment Consistent Equilibrium (CMCE) as an equilibrium in which the population regression coefficient in agents' perceived linear model matches the projection of actual returns onto the predictor.

Definition 1 (Cross-Moment Consistent Equilibrium) *A pair (μ, β) is a* Cross-Moment Consistent Equilibrium (CMCE) *if:*

[3] We use the following timing: at time t the agent forecasts $r_{t+1}^e = x_t\beta_t$ which then implies a realization for P_t and subsequently r_{t+1}. Therefore, to avoid simultaneity issues, we assume that when computing the current estimate, the last available log-return observation is r_{t-1}.

1. *μ is a non-degenerate invariant probability measure for the stochastic process*

$$r_{t-1} = \eta_{t-1} + g_\beta(x_{t-2}) - g_\beta(x_{t-1}),$$

where $x_t \overset{iid}{\sim} \mathcal{N}(0, \sigma_x^2)$.

2. *The parameter β is the population OLS coefficient of r_{t-1} on x_{t-2} under the invariant measure μ, that is,*

$$\beta = \frac{\mathrm{Cov}_\mu(x_{t-2} r_{t-1})}{\mathrm{Var}_\mu(x_{t-2}^2)}$$

While a rational expectations equilibrium requires that agents' perceived law of motion coincides with the actual law of motion, a CMCE requires only that the population regression coefficient in agents' perceived linear model matches the projection of actual returns onto the predictor x_{t-2}. This weaker notion of consistency is appropriate when agents use misspecified linear models to forecast returns generated by a nonlinear process. Despite the misspecification, there is a statistical self-consistency between the empirical relationship agents perceive and the one implied by the ALM.

Proposition 1 *Under RLS learning there exists a unique CMCE given by $\beta = 0$. The CMCE is locally stable under learning for $\kappa \in (0, 1)$.*

Proof. In Appendix B.1.

Notice that $\kappa = \delta a^{-\ell}\phi$ is strictly positive for all admissible parameter values. The restriction $\kappa < 1$ imposes an upper bound on the combination of the discount factor, risk aversion, and dividend growth, and is required for the REE to be well-defined. This condition holds for standard parameter calibrations.

Proposition 1 shows that with infinite memory, learning eliminates perceived predictability: agents eventually accumulate enough data to recognize that returns are fundamentally unpredictable. However, this convergence relies critically on the assumption that all historical observations receive equal weight in estimation. In practice, investors may discount older data either because they believe the economic environment is changing or because they face cognitive or computational constraints on memory. We now examine how limited memory alters the learning outcome.

5 Weighted Least Squares and Constant Gain Learning

We saw that memory, in the sense of the weight assigned to past observations, plays a key role in determining the learning outcome. Under RLS learning with decreasing gain, agents eventually learn that returns are unpredictable. However, in practice agents may assign more weight to recent observations, either because they believe that the data-generating process is time-varying, or because they have limited memory capacity. To capture this, the second learning mechanism we consider is constant gain learning. This obtains by replacing the decreasing gain $1/t$ in RLS with a constant gain $g \in (0, 1)$, and as we show in Appendix A.2 constant gain corresponds to an on-line estimation of WLS (Weighted Least Squares) with geometrically declining weights.

$$\beta_t = \beta_{t-1} + \gamma M_{t-1}^{-1} x_{t-2}(r_{t-1} - x_{t-2}\beta_{t-1}). \tag{24}$$

$$M_t = M_{t-1} + \gamma \left(x_{t-1}^2 - M_{t-1}\right). \tag{25}$$

$$r_{t-1} = \eta_{t-1} + \log\Big(1 - \kappa e^{w \tanh(x_{t-2}\beta_{t-2}/w)}\Big) - \log\Big(1 - \kappa e^{w \tanh(x_{t-1}\beta_{t-1}/w)}\Big). \tag{26}$$

Given that the gain is not decreasing and that the updating depends on the stochastic term η_{t-1}, differently to RLS, the learning mechanism cannot converge to a fixed point. However, under suitable conditions and for small values of γ, and under the same stability condition $\kappa \in (0, 1)$, it can be shown that the learning dynamics will fluctuate around the CMCE. We formalize this in the following proposition.

Proposition 2 (Distribution under Constant Gain Learning) *Under Constant gain learning with parameter γ, and for $\kappa < 1$, for small values of γ the estimated parameter β approximately follows a normal distribution*

$$\beta_t \sim \mathcal{N}\left(\beta^*, \gamma \frac{\sigma_d^2(1-\kappa)}{2\sigma_x^2}\right), \tag{27}$$

where $\beta^* = 0$ is the cross-moment consistent equilibrium.

Proof. In Appendix B.2.

Proposition 2 characterizes the stochastic fluctuations in beliefs under finite memory. The variance of the estimated coefficient, $\gamma\sigma_d^2(2\sigma_x^2(1-\kappa))^{-1}$, is driven by three fundamental parameters.

First, the gain parameter γ controls memory length and directly amplifies variance. Higher γ means agents discount past data more aggressively, giving recent observations disproportionate influence. This amplifies the impact of sampling noise: a few unusual return realizations can substantially shift beliefs when historical evidence receives little weight.

Second, dividend volatility σ_d^2 determines the fundamental uncertainty in returns. Greater return volatility generates larger forecast errors, which translate into more volatile coefficient estimates as agents attempt to fit a predictive relationship to noisy data.

Third, predictor variance σ_x^2 acts as a stabilizing force. When the predictor varies widely, each observation provides more information about the forecasting relationship, reducing estimation uncertainty.

6 Lasso Regression

We model belief formation under LASSO learning by endowing agents with a *soft-thresholded least squares* learning rule. Concretely, agents first update a least squares estimate of the forecasting coefficient using recursive or constant-gain learning, and then apply a soft-thresholding operator before forming expectations. This mechanism preserves analytical tractability while capturing the key feature of penalized learning, namely the tendency to set coefficients exactly to zero when empirical signals are weak.

Our empirical motivation is closely related to LASSO regression, which augments the least squares objective with an ℓ_1 penalty and is widely used to promote sparsity in estimated models. Rather than solving the LASSO problem directly within the learning dynamics, we adopt the soft-thresholding rule as a reduced-form representation of penalized estimation. As shown in Appendix A.3, this procedure is exactly equivalent to LASSO in the univariate case when regressors are orthonormal, in which case the LASSO estimator admits the closed-form representation

$$\hat{\beta}_{\text{LASSO}} = S(\beta_{\text{OLS}}, \lambda) = \text{sign}(\beta_{\text{OLS}}) \max\{0, |\beta_{\text{OLS}}| - \lambda\} . \tag{28}$$

While exact orthonormality is unlikely to hold in practice, it is common to work with predictors that are uncorrelated and standardized. In such environments, soft-thresholding provides a close approximation to the LASSO solution and is routinely used in empirical applications. We therefore refer to this learning mechanism as *LASSO learning*, with the understanding that it implements penalized least squares through an analytically tractable thresholding rule.

Under this specification, learning dynamics can be studied using the same recursive framework developed for least squares learning. The unpenalized coefficient continues to satisfy the standard cross-moment consistency condition, while sparsity arises endogenously from the thresholding step. As we show below, this modification leaves the location of the cross-moment consistent equilibrium unchanged but alters the stochastic behavior of beliefs under constant-gain learning. Specifically, the estimate for β_{OLS} will be distributed as a normal distribution centered around the fixed point, with variance proportional to the gain parameter γ. This implies that for small λ sometimes the estimates will be large enough that the LASSO estimate is non-zero. Given the distribution of β_{OLS}, we have:

Proposition 3 (Probability of Non-Zero LASSO Estimate) *Under Weighted Least Squares learning with geometrically declining weights with parameter γ, and for $\kappa < 1$, for small values of γ the probability that the LASSO estimate is different from zero is given by*

$$\mathbb{P}(|\beta_{OLS}| > \lambda) = 1 - \left[\Phi\left(\frac{\lambda}{\sigma_{OLS}}\right) - \Phi\left(\frac{-\lambda}{\sigma_{OLS}}\right) \right] , \tag{29}$$

where $\sigma_{OLS} = \left(\gamma s_d^2 \left(2\sigma_x^2 \left(1 + \frac{\kappa}{1-\kappa} \right) \right)^{-1} \right)^{1/2}$.

Proof. In Appendix B.3.

It is worth claryifing now the mechanism through which return predictability arises in our model. Under constant-gain learning, the evolution of beliefs can be written in on-line form as

$$\beta_t = \beta_{t-1} + \gamma M_{t-1}^{-1} x_{t-2}(r_{t-1} - x_{t-2}\beta_{t-1}), \tag{30}$$

where the term $r_{t-1} - x_{t-2}\beta_{t-1}$ is the one-step-ahead forecasting error. Large changes in β_t therefore arise when the forecasting error is large, when the regressor realization $|x_{t-2}|$ is large, or when the local second-moment estimate M_{t-1} is small. Given that returns evolve according to the ALM (26), it is clear that even when predictors x and innovations η are i.i.d. and independent, realizations in which x_{t-2} and η_{t-1} are accidentally aligned in sign and magnitude generate large forecast errors and hence large belief revisions. Economically, this implies that the probability of a regressor being selected as predictive, is larger when it seems to agents that innovations to the fundamental dividend process are correlated with movements in the predictor.

6.1 Extension to Multiple Uncorrelated Predictors

Our theoretical analysis focuses on the univariate case for analytical tractability, but all main results extend naturally to the multivariate setting when predictors are uncorrelated. Consider a PLM with k predictors: $r_{t+1} = X_t^T\beta + u_{t+1}$, where $X_t \sim \mathcal{N}(0, \Sigma_X)$ and Σ_X is diagonal. In this case, the second-moment matrix $M_t = \mathbb{E}[X_t X_t^T]$ is also diagonal, which implies that the learning dynamics decompose into k independent univariate problems. Specifically, each coefficient β_j evolves according to

$$\beta_{j,t} = \beta_{j,t-1} + \gamma M_{j,t-1}^{-1} x_{j,t-2}\left(r_{t-1} - X_{t-2}^T\beta_{t-1}\right), \tag{31}$$

where $M_{j,t} = \mathbb{E}[x_{j,t}^2]$ is the j-th diagonal element. The cross-moment consistent equilibrium condition similarly decomposes: $\beta_j = T_j(\beta)$ for all j, where $T_j(\beta) = \mathrm{Cov}(r_{t-1}, x_{j,t-2})/\mathrm{Var}(x_{j,t-2})$. Under constant-gain learning with penalty parameter λ, each unpenalized coefficient is approximately normally distributed with mean zero and variance $\gamma\sigma_d^2/(2\sigma_{x_j}^2(1-\kappa))$. The soft-thresholding operator then applies component-wise, and the probability that predictor j is selected becomes

$$\mathbb{P}(|\beta_j^{\mathrm{OLS}}| > \lambda) = 1 - \left[\Phi\left(\frac{\lambda}{\sigma_{\mathrm{OLS},j}}\right) - \Phi\left(\frac{-\lambda}{\sigma_{\mathrm{OLS},j}}\right)\right], \tag{32}$$

where $\sigma_{\mathrm{OLS},j}^2 = \gamma\sigma_d^2/(2\sigma_{x_j}^2(1-\kappa))$. Since the β_j are mutually independent when predictors are uncorrelated, the expected number of selected predictors is simply $\mathbb{E}[\#\,\text{selected}] = \sum_{j=1}^k \mathbb{P}(|\beta_j^{\mathrm{OLS}}| > \lambda)$. This decomposition also extends to the stability analysis: the Jacobian of the mean dynamics becomes block-diagonal, with each block corresponding to a single predictor, and the SRA framework applies after vectorizing the parameter space as $\theta_t = (\beta_t, \mathrm{vec}(M_t))$. In summary, uncorrelated predictors preserve the key insight that sparse predictability arises from finite-sample fluctuations in individual coefficients, with each predictor independently subject to selection when its estimated relationship happens to exceed the penalty threshold.

7 Empirical Application

7.1 Data

To evaluate the model in a realistic environment, we test its predictions using a large cross-section of assets and a high-dimensional set of news-based predictors. Our sample consists of the daily returns of all stocks that were continously listed on the New York Stock Exchange (NYSE) between 2010 and 2017. As predictors, we use the 180 daily news topic attention series constructed by Bybee et al. (2024) from the full text of the *Wall Street Journal* using an unsupervised Latent Dirichlet Allocation (LDA) model. The LDA assigns each article a vector of topic proportions, which Bybee et al. (2024) aggregate to the daily frequency to measure media attention across a broad set of business, economic, political, and social topics. A complete list of all 180 topics is provided in Table 4. The resulting predictor set is highly heterogeneous, with little theoretical guidance as to which variables are relevant. In this environment, it is natural for agents to rely on LASSO-based learning to identify the most informative signals.

We use innovations in topic attention rather than levels, as media attention is highly persistent and levels primarily capture slow moving components of news coverage. To isolate unexpected changes in information, we estimate an AR(1) process for each topic attention series and define X_t as the corresponding residuals. Consistent with the model, we work with log returns, defined as $r_{i,t} = \log(R_{i,t})$, where $R_{i,t}$ denotes the gross return of stock i on day t. The final sample contains returns of 1,620 stocks and news attention to 180 topics over 1,889 trading days. Table 1 reports summary statistics of our data.

Table 1: Summary Statistics

Variable	Mean	Std. Dev.	Min	Max	Skewness	Kurtosis	N
Log return	0.000	0.021	-1.964	2.051	-0.587	101.917	3,060,180
Topic level	0.006	0.004	0.000	0.116	3.662	33.361	340,020
Topic innovation	0.000	0.003	-0.054	0.062	1.885	16.019	339,840

Note: The table reports summary statistics for daily log returns $r_{i,t} = \log(R_{i,t})$ of NYSE-listed stocks from 2010–2017 and for news-topic attention measures constructed following Bybee et al. (2024). Topic attention innovations are defined as residuals from topic-specific AR(1) regressions estimated over the full sample.

7.2 Estimation Strategy

Our theory has two core components: (i) a misspecified PLM through which agents form beliefs about return predictability, and (ii) an ALM that maps these beliefs into realized returns. Because neither the agents nor the econometrician observe the true data-generating process for returns, our empirical strategy mirrors the agents' learning problem. We begin by estimating the PLM as in the model, while introducing some modifications to make the framework empirically implementable.

First, we assume that agents estimate the PLM using a rolling window of fixed length rather than geometrically declining weights, which is more in line with standard econometric practices when dealing with potential time-varying parameters and allow us to closely mirror the design of Chinco, Clark-Joseph and Ye (2019). Second, we endow agents with a full LASSO regression rather than the soft-thresholding approximation used in the theoretical analysis. As already remarked, the soft-thresholding operator is exactly equivalent to LASSO when regressors are orthonormal. Instead, we allow regressors to be approximatly uncorrelated, which is a weaker but more realistic condition. Third, we allow for lags of the regressors to enter the PLM, by simply stacking n lags of each topic innovation as potential predictors. This extension captures the possibility that agents consider delayed effects of news topics on returns. Accordingly, equation (28) need not hold exactly, but should provide a close approximation. Consistent with this assumption, Figure 3 in Appendix C shows that topic-attention innovations are close to contemporaneously uncorrelated, while Table 5 documents that cross-lag correlations in the time series are also small. At each date t, the information set consists of k topic-attention innovations. Let $x_t \in \mathbb{R}^k$ denote the vector of innovations at time t, with element $x_{j,t}$ corresponding to topic $j \in \{1, \dots, k\}$. For a rolling window of length s, define the regressor matrix by stacking past topic vectors row-wise,

$$X_t \in \mathbb{R}^{s\times k} \equiv \begin{bmatrix} x_{t-s}^\top \\ x_{t-s+1}^\top \\ \vdots \\ x_{t-1}^\top \end{bmatrix}, \qquad r_{i,t+1} \in \mathbb{R}^s \equiv \begin{bmatrix} r_{i,t-s+1} \\ r_{i,t-s+2} \\ \vdots \\ r_{i,t} \end{bmatrix}. \tag{33}$$

Each row of X_t is a $1 \times k$ vector of topic innovations observed at a given date, while each column tracks the time series of innovations for a given topic over the window.

For each stock i and each window ending at $t-1$, the agent estimates a LASSO coefficient vector

$$\beta_{i,t} \in \mathbb{R}^k, \tag{34}$$

so that beliefs are indexed by both time and topic. Equivalently, the collection $\{\beta_{i,t}\}_t$ defines a stock-specific, time-varying belief process, where the j-th element $\beta_{i,t}^{(j)}$ measures the perceived predictive content of topic j for stock i at date t. The empirical perceived law of motion (PLM) for log returns is therefore

$$r_{i,t+1} = X_t \beta_{i,t} + \eta_{i,t+1}, \tag{35}$$

where $X_t\beta_{i,t} \in \mathbb{R}^s$ denotes the vector of fitted within-window returns implied by the k-dimensional belief vector at date t. Applying this procedure sequentially yields a time-varying sequence of belief parameters $\{\beta_{i,t}\}$ for each stock.

Using the estimated belief path, we then estimate the empirical actual law of motion

(ALM) implied by the asset-pricing equation.[4]

$$r_{i,t+1} = c_i + \log\Big(1 - \kappa_i e^{x_t^\top \beta_{i,t}}\Big) - \log\Big(1 - \kappa_i e^{x_{t+1}^\top \beta_{i,t+1}}\Big) + \eta_{i,t+1}, \tag{36}$$

where $x_t^\top \beta_{i,t}$ is scalar because $x_t, \beta_{i,t} \in \mathbb{R}^k$. The stock-specific parameter κ_i is estimated by nonlinear least squares.

Before implementing our estimation strategy, we calibrate the three parameters governing the agent's learning rule: the rolling window length s, the penalty parameter λ, and the number of predictor lags n considered. To do this we operate at market level, estimating the PLM and ALM using the CRSP value-weighted market return as the forecasting target and the full set of topic-innovation series as potential signals. We select these parameters using Bayesian optimization to maximize an objective function which is a combination of first and second-stage fit as

$$\text{Objective}(s, \lambda, n) = \kappa \cdot R^2_{\text{ALM}} \cdot \mathbb{I}\left(R^2_{\text{PLM}} > 0\right) \cdot \mathbb{I}\left(\kappa_{t-stat} > 1.96\right)$$

where R^2 is the in-sample R^2 of the corresponding stage, κ_{t-stat} is the t-statistic of the NLS κ coefficient.

The objective is chosen to disciplinee the empirical implementation by restricting attention to learning rules under which the PLM-ALM structure is well defined.

The term $\mathbb{I}(R^2_{\text{PLM}} > 0)$ serves two purposes. First, it rules out parameterizations for which beliefs contain no predictive content, in which case the belief channel in the ALM is unidentified. Second, it provides a minimal empirical justification for why agents would rely on a LASSO-based forecasting rule in the first place. If the perceived law of motion delivers no explanatory power, there would be no reason for agents to condition their decisions on the estimated coefficients $\beta_{i,t}$.

Conditional on a nondegenerate PLM, maximizing R^2_{ALM} selects configurations in which belief dynamics translate into economically relevant price movements. Finally, imposing $\mathbb{I}(\kappa_{t\text{-stat}} > 1.96)$ ensures that the key nonlinear feedback parameter is statistically identified.

Lastly, this procedure allows us to choose hyper-parameters once, at the market level and then keep them fixed when estimating the model for individual stocks. In this way we can ensure that cross-sectional variation is not driven by different hyper-parameter choices. The optimization yields the following calibration: rolling window length $s = 320$ days, penalty parameter $\lambda = 0.0025$, and number of predictor lags $n = 21$.

7.3 Results

Having calibrated the learning parameters, we now implement the model in the cross-section. For each stock, we estimate the PLM in equation (35) using a rolling-window LASSO. This procedure delivers a time series of belief coefficients $\{\beta_{i,t}\}$ and corresponding

[4]While the theoretical model implies zero unconditional mean log returns under our maintained assumptions, the empirical specification includes an intercept to absorb average return components not captured by the model.

one-step-ahead forecasts, which we summarize in Table 2.

Table 2: Summary Statistics of LASSO Estimation Results

	mean	std	median	min	max	p05	p95
In-sample R^2	0.1056	0.1149	0.0756	-0.0009	0.5041	-0.0007	0.3161
Out-of-sample R^2	-0.0231	0.0344	-0.0116	-0.1788	0.0037	-0.0766	-0.0003

Note: The table reports summary statistics across stocks for the first-stage LASSO in-sample R^2, computed within each rolling estimation window and then averaged over windows, and the first-stage out-of-sample R^2, computed from the sequence of one-step-ahead LASSO forecasts evaluated against realized (demeaned) returns with the zero-forecast as benchmark.

As we can see in Table 2 the in-sample R^2 is positive, indicating that the LASSO learning rule explains returns within the estimation window, but not outside of it as the out-of-sample R^2 is negative on average. This pattern is consistent with our theory: agents perceive return predictability in-sample, but this predictability does not persist out-of-sample due to the misspecified nature of the PLM and the high-dimensional environment.

Next, we document that perceived predictability is concentrated in a few signals when it arises. Figure 1 plots the average number of selected predictors per stock over time, as well as shaded areas corresponding to the 5th and 95th percentiles across stocks. Out of the whole possible 3780 regressors, the number of selected predictors is small, indicating that when agents believe in predictability, they typically attribute it to only a small subset of available signals.

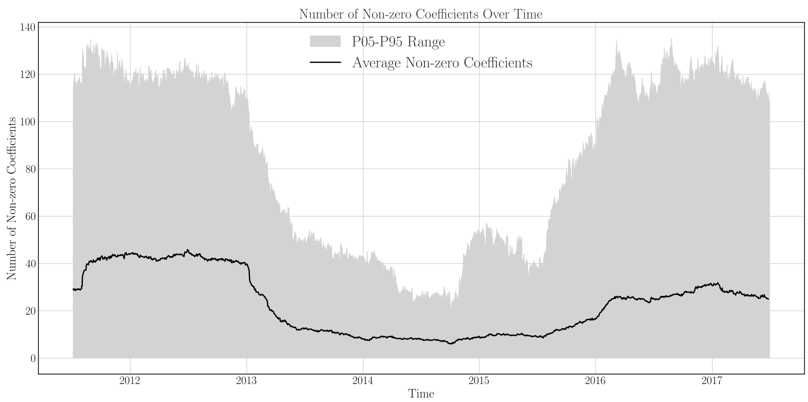

Figure 1: Average Number of Active Coefficients over Time

Finally, we test the cross-sectional implication in Proposition 3 that more volatile stocks should exhibit more frequent selection. Figure 2 plots, for each stock, the average number of non-zero coefficients selected per window against the variance of its returns. The relationship is clearly positive: stocks with higher return variance display more active coefficients on average.

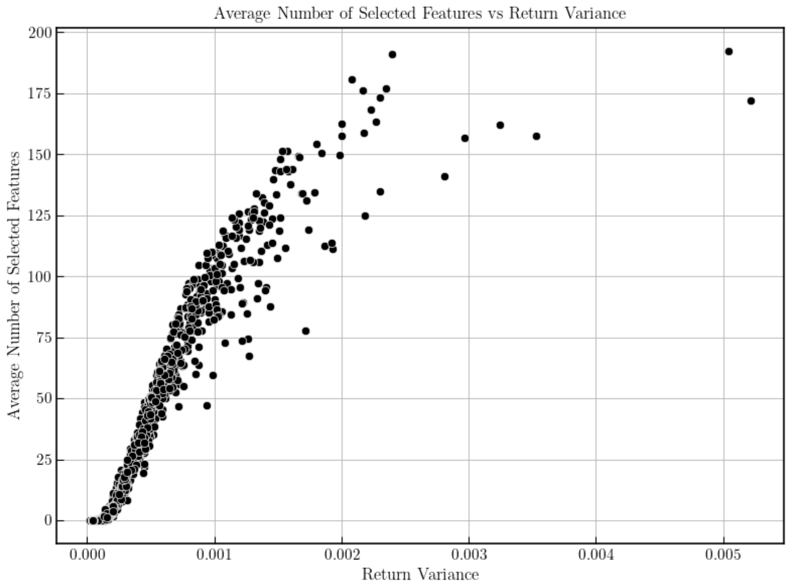

Figure 2: Average Number of Non-Zero Coefficients vs. Return Variance

We view these patterns as evidence in favor of our proposed learning rule. An agent applying LASSO to the data would find the rule attractive ex ante, because it delivers non-trivial in sample fit, even though its out of sample performance is weak. Consistent with our theory, perceived predictability is episodic, so at any point in time only a subset of stocks is viewed as predictable and, conditional on selection, beliefs are sparse and concentrated on a few predictors. Taken together, these results match the model's predictions and mirror recent evidence that return predictability arises in short lived, shifting pockets rather than as a stable forecasting relation (Chinco, Clark-Joseph and Ye, 2019).

Next, we estimate the parameter κ_i implied by the ALM by applying NLS to equation (36), taking the LASSO-estimated belief sequence $\beta_{i,t}$ as given. Identification relies on time-series variation in agents' beliefs about return predictability, as captured by fluctuations in $X_t\beta_{i,t}$, and on the corresponding variation in realized returns. Intuitively, κ_i is pinned down by how changes in expected returns translate into realized returns through the nonlinear asset-pricing equation.Table 3 shows the summary of this estimation for the cross-section of stocks. A significant κ indicates that the non-linear ALM has explanatory power over a simple case where returns are unpredictable, and we view it as an empirical test of the CMCE equilibrium being at work in the data. The proportion of stocks for which the κ estimate is statistically significant at least at the 5% level is 25%. For these stocks, we summarize the estimation results in Table 3.

Table 3: Second-Stage Estimation: Summary Statistics

	mean	std	median	min	max	p05	p95
R^2	0.0016	0.0011	0.0016	0.0001	0.0038	0.0002	0.0033
$\hat{\kappa}$	0.6435	0.2718	0.6637	0.1311	0.9711	0.2337	0.9683
t-statistic of $\hat{\kappa}$	9.5058	12.7404	4.7980	1.9954	44.1949	2.0958	34.4050

Note: The table reports summary statistics across stocks for second-stage predictive performance and belief-implied return sensitivity. $\hat{\kappa}$ denotes the estimated sensitivity parameter and its t-statistic. Each stock-level estimation uses 1,500 daily observations, and the reported statistics are computed across the full cross-section of 1,620 stocks.

8 Conclusion

This paper develops an asset-pricing model in which investors search for return predictability in a high-dimensional information environment using a LASSO-based learning rule. Agents estimate a penalized linear forecasting model, form expectations based on the selected predictors, and these beliefs feed back into prices through a nonlinear asset-pricing equation. Because perceived and actual laws of motion differ, learning does not converge to rational expectations; instead, the economy settles into a cross-moment consistent equilibrium (CMCE). The central implication is that the act of searching for predictability can itself generate sparse and short-lived windows of return predictability. In finite samples, agents occasionally select predictors by chance. When this occurs, trading on the perceived signal feeds back into prices, making the predictability temporarily self-fulfilling. As new data arrive and older observations drop out of the agents' effective memory, coefficient estimates revert back to zero, and the induced predictability disappears. The theory yields tractable conditions linking the frequency of selection events to fundamentals and features of the information environment. Empirically, we implement the model on daily returns for 1,620 NYSE-listed stocks from 2010–2017 using 180 Wall Street Journal topic-attention innovations as predictors. Consistent with the model, predictability is episodic and sparse, and selection occurs more frequently for higher-volatility stocks.

We plan to extend our analysis along several dimensions. The current model features a representative agent; a natural next step is to introduce heterogeneity in learning behavior. On the empirical side, the model cannot explain heterogeneity in the probability that different features are selected, beyond what is implied by their statistical properties. The data, however, exhibit systematic cross-feature variation in selection frequencies. We therefore test whether this variation reflects an additional, non-statistical component of behavior. Specifically, we hypothesize that, after estimating the statistical model, agents are more likely to act on its signals when the resulting predictions are consistent with their pre-existing narratives about how the stock market works and about plausible causal links between specific features and asset prices.

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A Derivation of Recursive Learning Models

A.1 Recursive Least Squares

The first updating mechanism we consider is Recursive Least Squares (RLS), which is a recursive implementation of the standard Ordinary Least Squares (OLS). In OLS, the agent estimates the coefficients of a linear regression model by minimizing the sum of squared residuals. Assuming agents use OLS to estimate the parameters in the PLM (10), the OLS estimator β_t can be derived from the minimization[5] problem as

$$\beta_t = \left(\sum_{i=1}^{t-1} x_{i-1}^2 \right)^{-1} \sum_{i=1}^{t-1} x_{i-1} r_i. \tag{37}$$

Estimating OLS on the whole available data set is sometimes referred to as off-line estimation. Another approach is to estimate the model on-line, that is, as soon as a new data point arrives. It turns out that the estimator formula allows for a recursive formulation. Define

$$M_{t-1} = \frac{1}{t-1} \left(\sum_{i=1}^{t-1} x_{i-1}^2 \right), \quad \beta_t = \frac{1}{t-1} M_{t-1}^{-1} \sum_{i=1}^{t-1} x_{i-1} r_i. \tag{38}$$

First we establish a recursion for M_t. Start from

$$\frac{t-1}{t} M_{t-1} = \frac{1}{t} \left(\sum_{i=1}^{t-1} x_{i-1}^2 \right) \implies \frac{t-1}{t} M_{t-1} + \frac{1}{t} x_{t-1}^2 = M_t, \tag{39}$$

that is

$$M_t = M_{t-1} + \frac{1}{t} \left(x_{t-1}^2 - M_{t-1} \right). \tag{40}$$

For β_t we have

$$\begin{aligned} \beta_t &= \beta_{t-1} + [(t-1)M_{t-1}]^{-1} \{x_{t-2} r_{t-1} + [(t-2)M_{t-2}] \beta_{t-1}\} - \beta_{t-1} \\ &= \beta_{t-1} + [(t-1)M_{t-1}]^{-1} \left\{x_{t-2} r_{t-1} - x_{t-2}^2 \beta_{t-1}\right\}. \end{aligned} \tag{41}$$

A.2 Constant Gain Learning

In Weighted Least Squares (WLS), the agent estimates the coefficients of a linear regression model by minimizing a weighted sum of squared residuals. The weights are typically chosen to reflect the relative importance or reliability of different observations. Assuming the linear model (10), the WLS estimator can be derived from the minimization problem

$$\min_{\beta} \sum_{i=1}^{t-1} w_{t-1,i} (r_i - \beta x_{i-1})^2, \tag{42}$$

with solution

$$\beta = \left(\sum_{i=1}^{t-1} w_{t-1,i} x_{i-1}^2 \right)^{-1} \sum_{i=1}^{t-1} w_{t-1,i} x_{i-1} r_i. \tag{43}$$

In particular, we are interested in the case where the weights decline geometrically over

[5] We use the following timing: at time t the agent forecasts $r_{t+1}^e = x_t \beta_t$ which then implies a realization for r_t. Therefore, to avoid simultaneity issues, we assume that when computing the current estimate, the last available log-return observation is r_{t-1}.

time[6]

$$w_{t,i} = \gamma(1-\gamma)^{t-i}, \quad \gamma \in (0,1). \tag{44}$$

Similar to the OLS case, a recursive relationship can be derived.

Define

$$M_{t-1} = \gamma \sum_{i=1}^{t-1} (1-\gamma)^{t-1-i} x_{i-1}^2, \quad \beta_t = \gamma M_{t-1}^{-1} \sum_{i=1}^{t-1} (1-\gamma)^{t-1-i} x_{i-1} r_i. \tag{45}$$

First we establish a recursion for M_t. Start from

$$M_{t-1} = \gamma \sum_{i=1}^{t-1} (1-\gamma)^{t-1-i} x_{i-1}^2, \tag{46}$$

then

$$(1-\gamma) M_{t-1} = \gamma \sum_{i=1}^{t-1} (1-\gamma)^{t-i} x_{i-1}^2 = M_t - \gamma x_{t-1}^2. \tag{47}$$

That is

$$M_t = M_{t-1} + \gamma \left(x_{t-1}^2 - M_{t-1} \right). \tag{48}$$

Similarly for β_t we start from the lagged expression

$$\beta_{t-1} = \gamma M_{t-2}^{-1} \sum_{i=1}^{t-2} (1-\gamma)^{t-2-i} x_{i-1} r_i, \tag{49}$$

then

$$(1-\gamma)\beta_{t-1} = \gamma M_{t-2}^{-1} \sum_{i=1}^{t-1} (1-\gamma)^{t-1-i} x_{i-1} r_i, \tag{50}$$

and

$$\beta_{t-1}(M_{t-1} - \gamma x_{t-2}^2) = \gamma \sum_{i=1}^{t-1} (1-\gamma)^{t-1-i} x_{i-1} r_i \to \beta_{t-1}(1 - \gamma M_{t-1}^{-1} x_{t-2}^2) = \gamma M_{t-1}^{-1} \sum_{i=1}^{t-1} (1-\gamma)^{t-1-i} x_{i-1} r_i, \tag{51}$$

since

$$M_{t-2} = \frac{1}{1-\gamma}(M_{t-1} - \gamma x_{t-2}^2).$$

Adding $\gamma M_{t-1}^{-1} x_{t-2} r_{t-1}$ to both sides we obtain

$$\beta_{t-1}(1 - \gamma M_{t-1}^{-1} x_{t-2}^2) + \gamma M_{t-1}^{-1} x_{t-2} r_{t-1} = \beta_t, \tag{52}$$

[6] In empirical applications, instead of using geometrically declining weights, it is common to use a rolling window approach, where only the most recent observations within a fixed-size window are used for estimation. However, the latter is analytically intractable. It is possible to determine a relationship between the window size and the geometric decay parameter γ such that the two methods give similar weights *overall* on the ℓ most recent observations: $\ell \approx \log(1-p)/\log(1-\gamma)$, where p is the proportion of total weight assigned to the ℓ most recent observations.

which can be rearranged as

$$\beta_t = \beta_{t-1} + \gamma M_{t-1}^{-1} x_{t-2}(r_{t-1} - x_{t-2}\beta_{t-1}). \tag{53}$$

A.3 Lasso Learning

The LASSO (Least Absolute Shrinkage and Selection Operator) estimator is defined as the solution to the optimization problem

$$\hat{\beta}_{\text{LASSO}} = \arg\min_{\beta} \left\{ \frac{1}{2}\left[(Y - X\beta)'(Y - X\beta)\right] + \lambda \sum_{j=1}^{k} |\beta_j| \right\},$$

where $\lambda \geq 0$ is a tuning parameter that controls the amount of regularization. The LASSO estimator can be interpreted as a trade-off between fitting the data well (minimizing the sum of squared residuals) and keeping the model simple (minimizing the absolute values of the coefficients).

In general there is no closed form solution for the LASSO estimator, since the absolute value function is not differentiable at zero. However, in the special case where X is an orthonormal matrix, the LASSO estimator can be expressed in closed form. To do so we can rewrite the optimization problem as

$$\hat{\beta}_{\text{LASSO}} = \arg\min_{\beta} \left\{ \frac{1}{2} \left[Y'Y - 2\beta' \underbrace{X'Y}_{\beta_{\text{OLS}}} + \beta' \underbrace{X'X}_{I} \beta \right] + \lambda \sum_{j=1}^{k} |\beta_j| \right\},$$

where the underbrackets follow from orthonormality and notice that the term $Y'Y$ does not depend on β, so it does not change the minimizer. Using $A'A = \sum_{j=1}^{k} a_j^2$, we can rewrite the problem as

$$\hat{\beta}_{\text{LASSO}} = \arg\min_{\beta} \left\{ \sum_{j=1}^{k} \left(-\beta_j \beta_{\text{OLS},j} + \frac{1}{2}\beta_j^2 + \lambda |\beta_j| \right) \right\}.$$

This problem is separable in the coefficients β_j, so we can solve for each coefficient independently. The optimization problem for each j coefficient is

$$\min_{\beta_j} \left\{ \frac{1}{2}\beta_j^2 - \beta_j \beta_{\text{OLS},j} + \lambda |\beta_j| \right\}.$$

To solve this problem, we consider two cases:

1. $\beta_{\text{OLS},j} > 0$, then β_j has to be positive, otherwise we could decrease the objective by setting $\beta_j = 0$. In this case, the problem becomes

$$\min_{\beta_j \geq 0} \left\{ \frac{1}{2}\beta_j^2 - \beta_j \beta_{\text{OLS},j} + \lambda \beta_j \right\},$$

and differentiating and setting to zero gives

$$\beta_j - \beta_{\text{OLS},j} + \lambda = 0 \Rightarrow \beta_j = \beta_{\text{OLS},j} - \lambda.$$

Imposing non-negativity yields

$$\hat{\beta}_{\text{LASSO},j} = \max\left(0, \beta_{\text{OLS},j} - \lambda\right).$$

2. $\beta_{\text{OLS},j} < 0$, then β_j has to be negative, otherwise we could decrease the objective by setting $\beta_j = 0$. In this case, the problem becomes

$$\min_{\beta_j < 0} \left\{ \frac{1}{2}\beta_j^2 - \beta_j \beta_{\text{OLS},j} + \lambda(-\beta_j) \right\},$$

and

$$\beta_j - \beta_{\text{OLS},j} - \lambda = 0 \Rightarrow \beta_j = \beta_{\text{OLS},j} + \lambda.$$

Imposing negativity yields

$$\hat{\beta}_{\text{LASSO},j} = \min\left(0, \beta_{\text{OLS},j} + \lambda\right).$$

Combing the two cases, we obtain

$$\hat{\beta}_{\text{LASSO}} = S(\beta_{\text{OLS}}, \lambda) = \text{sign}(\beta_{\text{OLS}}) \cdot \max\left(0, |\beta_{\text{OLS}}| - \lambda\right), \tag{54}$$

where $S(\cdot)$ is the soft-thresholding operator.

B Proofs

B.1 Proof of proposition 1

In a CMCE the value of β solves

$$T(\beta) - \beta = 0, \tag{55}$$

where

$$T(\beta) = \frac{\text{Cov}(r_{t-1}, x_{t-2})}{\text{Var}(x_{t-2})} = \frac{\text{Cov}(r_{t-1}, x_{t-2})}{\sigma_x^2}, \tag{56}$$

and r_{t-1} is given by (23) evaluated at the stationary distribution.

To compute $\text{Cov}(r_{t-1}, x_{t-2})$ we exploit the fact that x_{t-2} is independent of η_{t-1}^d and x_{t-1}, which implies that

$$\text{Cov}(r_{t-1}, x_{t-2}) = \text{Cov}\left(g_\beta(x_{t-2}), x_{t-2}\right). \tag{57}$$

By Stein's lemma

$$\text{Cov}(r_{t-1}, x_{t-2}) = \text{Cov}(g_\beta(x_{t-2}), x_{t-2}) = \sigma_x^2 \mathbb{E}\left[g'_\beta(x_{t-2})\right]. \tag{58}$$

This implies that $T(\beta) = \mathbb{E}\left[g'_\beta(x_{t-2})\right]$.

We can show that $\operatorname{sign}[T(\beta)] = -\operatorname{sign}(\beta)$ unless $\beta = 0$. In fact, we have

$$g'_\beta(x) = -\frac{\kappa\,\beta\,e^{w\tanh(x\beta/w)}\,\operatorname{sech}^2(x\beta/w)}{1-\kappa e^{w\tanh(x\beta/w)}}. \tag{59}$$

In the numerator, every multiplicative factor except β is strictly positive since $\kappa > 0$, $e^{w\tanh(x\beta/w)} > 0$, and $\operatorname{sech}^2(x\beta/w) > 0$. The denominator is also strictly positive since w is chosen exactly to make the second term smaller than 1. Hence the sign of $g'_\beta(x)$ is determined entirely by $-\beta$. In particular,

$$\operatorname{sign}[T(\beta)] = \operatorname{sign}\left[-\beta\,\mathbb{E}\left(\frac{\kappa\,e^{w\tanh(x\beta/w)}\,\operatorname{sech}^2(x\beta/w)}{1-\kappa e^{w\tanh(x\beta/w)}}\right)\right] = -\operatorname{sign}(\beta). \tag{60}$$

Therefore $T(\beta)$ has the opposite sign of β for all $\beta \neq 0$, and the only fixed point is $\beta = 0$.

To study local stability of the CMCE under RLS learning we follow the approach in Chapter 6 of Evans and Honkapohja (2001), which casts the learning dynamics as a Stochastic Recursion Algorithm (SRA). The key technical issue is that the return innovation r_{t-1} depends on both β_{t-1} and β_{t-2}. To deal with this, we keep the *parameter* vector first order and move the required lags into the *state* vector. Specifically, define the parameter vector as

$$\theta_t \equiv \begin{pmatrix}\beta_t \\ M_t\end{pmatrix},$$

with domain $D = \mathbb{R} \times (0,\infty)$, and the gain sequence $\gamma_t = t^{-1}$.

We augment the state to include the lagged belief needed by the ALM:

$$Z_t \equiv \begin{pmatrix} x_{t-1} \\ x_{t-2} \\ \eta_{t-1} \\ s_{t-1} \\ s_{t-2}\end{pmatrix}, \qquad s_{t-1} \equiv \beta_{t-1}.$$

Let $W_t = (x_{t-1}, \eta_{t-1}, 1)'$. Then the state follows a conditionally linear law of motion of the form

$$Z_t = AZ_{t-1} + B(\theta_{t-1})W_t, \quad A = \begin{pmatrix} 0&0&0&0&0\\ 1&0&0&0&0\\ 0&0&0&0&0\\ 0&0&0&0&0\\ 0&0&0&1&0\end{pmatrix}, \quad B(\theta) = \begin{pmatrix}1&0&0\\0&0&0\\0&1&0\\0&0&\beta\\0&0&0\end{pmatrix}.$$

This construction ensures $s_{t-1} = \beta_{t-1}$ in real time, while under a fixed parameter $\theta = (\beta, M)$ we have $s \equiv \beta$ in the associated frozen-parameter state process $\bar{Z}_t(\theta)$. Next, write the parameter recursion in SRA form as

$$\theta_t = \theta_{t-1} + \gamma_t \, \mathcal{H}(\theta_{t-1}, Z_t) + \gamma_t^2 \, \rho_t(\theta_{t-1}, Z_t),$$

by rearranging (41) to get

$$\begin{aligned} \beta_t &= \beta_{t-1} + t^{-1} M_{t-1}^{-1} \left[x_{t-2} r_{t-1} - x_{t-2}^2 \beta_{t-1} \right] \\ &\quad + t^{-2} \frac{t}{t-1} M_{t-1}^{-1} \left[x_{t-2} r_{t-1} - x_{t-2}^2 \beta_{t-2} \right]. \end{aligned} \tag{61}$$

and β_{t-2} is replaced by the state component s_{t-2}, so that the required functions are

$$\mathcal{H}(\theta, Z) = \begin{pmatrix} M^{-1}[x_2 \, r(\beta, s_2; x_1, x_2, \eta) - x_2^2 \, \beta] \\ x_1^2 - M \end{pmatrix}, \qquad \rho_t(\theta, Z) = \begin{pmatrix} \frac{t}{t-1} \, M^{-1}[x_2 \, r(\beta, s_2; x_1, x_2, \eta) - x_2^2 \, s_2] \\ 0 \end{pmatrix},$$

with $Z = (x_1, x_2, \eta, s_1, s_2)$ and

$$r(\beta, s_2; x_1, x_2, \eta) = \eta + \log\Big(1 - \kappa e^{w \tanh(x_2 s_2 / w)}\Big) - \log\Big(1 - \kappa e^{w \tanh(x_1 \beta / w)}\Big).$$

Assumptions (A.1)–(A.3) and (B.1)–(B.2) in Section 6.2.1 of Evans and Honkapohja (2001) hold on any compact $Q \subset D$: (A.1) holds because $\gamma_t = t^{-1}$ implies $\sum_t \gamma_t = \infty$ and $\sum_t \gamma_t^2 < \infty$; (B.1) holds because W_t is i.i.d. with finite moments of all orders under $\eta_{t-1} \sim \mathcal{N}(0, \sigma^2)$ and $x_t \sim \mathcal{N}(0, \sigma_x^2)$; (B.2) holds since A is constant with spectral radius $0 < 1$ and $B(\theta)$ is uniformly bounded and Lipschitz on Q; (A.2) holds because $\tanh(\cdot)$ and $\operatorname{sech}(\cdot)$ are bounded and, under $\kappa e^w < 1$, the term $\log(1 - \kappa e^{w \tanh(\cdot)})$ is uniformly bounded away from its singularity on Q, implying polynomial growth of $\mathcal{H}$ and ρ_t in (x_1, x_2, ℓ), whose moments are finite; (A.3) follows because $\mathcal{H}$ is C^2 in θ on Q with bounded second derivatives (again using bounded derivatives of $\tanh$ and the restriction $\kappa e^w < 1$), hence satisfies the required Lipschitz properties. Therefore the associated ODE is

$$\dot{\theta} = h(\theta), \qquad h(\theta) = \lim_{t \to \infty} \mathbb{E}[\mathcal{H}(\theta, \bar{Z}_t(\theta))].$$

Under fixed $\theta = (\beta, M)$, the frozen state is stationary with $x_1, x_2 \overset{iid}{\sim} \mathcal{N}(0, \sigma_x^2)$, $\eta \sim \mathcal{N}(0, \sigma_d^2)$ and $s \equiv \beta$, so that

$$h(\beta, M) = \begin{pmatrix} M^{-1}(\mathbb{E}[x_2 r(\beta, \beta; x_1, x_2, \ell)] - \beta \sigma_x^2) \\ \sigma_x^2 - M \end{pmatrix} = \begin{pmatrix} (\sigma_x^2 / M)(T(\beta) - \beta) \\ \sigma_x^2 - M \end{pmatrix},$$

where $\mathbb{E}[x_2 r(\beta, \beta; \cdot)] = \sigma_x^2 T(\beta)$ by the definition of $T(\beta)$. The CMCE corresponds to the unique fixed point $(\beta^*, M^*) = (0, \sigma_x^2)$. Linearizing h at $(0, \sigma_x^2)$ yields Jacobian

$$Dh(0, \sigma_x^2) = \begin{pmatrix} T'(0) - 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Moreover, from $g'_\beta(x) = -(\kappa \beta e^{w \tanh(x\beta/w)} \operatorname{sech}^2(x\beta/w))/(1 - \kappa e^{w \tanh(x\beta/w)})$ we obtain

$T(\beta) = \mathbb{E}[g'_\beta(x)]$ and hence $T'(0) = -(\kappa/(1-\kappa))$, so $T'(0) - 1 = -1/(1-\kappa) < 0$ for all $\kappa \in (0,1)$. Thus $(0, \sigma_x^2)$ is locally asymptotically stable for the ODE, and by the E-stability principle it is also locally stable under RLS learning.

B.2 Proof of proposition 2

For the proof we use the small-γ Gaussian approximation following Chapter 7 in Evans and Honkapohja (2001). We use the same state-augmentation device as above to eliminate the apparent second-order dependence, and all objects are defined as in the RLS case, except that now the gain sequence is constant: $\gamma_t \equiv \gamma \in (0,1)$. To verify the assumptions for the Gaussian approximation, we work on an open domain $D = \{(\beta, M) : \beta \in \mathbb{R},\ M \in (\zeta, \infty)\}$ with $\zeta > 0$ and (as standard in constant-gain RLS) impose a projection onto an arbitrary compact $Q \Subset D$ to guarantee $M_t \geq \zeta$; on Q, (A.2) and (A.3′) hold because $H(\theta, Z)$ is C^1 in (θ, Z) with polynomial growth in (x_1, x_2, η), and $\tanh(\cdot)$ together with $\kappa e^w < 1$ keep the logarithmic term uniformly away from its singularity, implying local Lipschitz and polynomial bounds for H and $\partial H/\partial x$; (M.1)–(M.5) and (N.1)(i)–(iii) hold because the augmented state is a geometrically ergodic Markov chain under fixed θ (linear shift with i.i.d. Gaussian innovations) and has finite moments of all orders; (H.1) holds since $h(\theta) \equiv \mathbb{E}[H(\theta, Z^\theta)]$ has continuous first and second derivatives on D; (H.2)–(H.3) hold locally around the CMCE because the Jacobian $B \equiv D_\theta h(\theta^*)$ has eigenvalues with strictly negative real parts (computed below), and $D_\theta h$ is Lipschitz on Q. Therefore Theorem 7.9 applies, yielding for small γ and large t the approximation $\theta_t \approx N(\theta^*, \gamma C)$ with

$$C = \int_0^\infty e^{sB} R(\theta^*) e^{sB'}\, ds, \qquad R_{ij}(\theta) = \sum_{k=-\infty}^{\infty} \operatorname{cov}(H_i(\theta, Z_k^\theta), H_j(\theta, Z_0^\theta)),$$

where Z_k^θ denotes the stationary frozen-θ chain. In our case, under frozen $\theta = (\beta, M)$ we have

$$h(\beta, M) = \begin{pmatrix} (\sigma_x^2/M)(T(\beta) - \beta) \\ \sigma_x^2 - M \end{pmatrix},$$

so the unique equilibrium is $\theta^* = (\beta^*, M^*) = (0, \sigma_x^2)$ and

$$B = D_\theta h(\theta^*) = \begin{pmatrix} T'(0) - 1 & 0 \\ 0 & -1 \end{pmatrix} = \begin{pmatrix} -\frac{1}{1-\kappa} & 0 \\ 0 & -1 \end{pmatrix},$$

where $T'(0) = -(\kappa/(1-\kappa))$ as in the decreasing-gain analysis. At θ^*, $r(\beta, s; \cdot) = \eta$ because $\tanh(0) = 0$ and the two log-terms cancel, hence

$$H_1(\theta^*, Z) = \sigma_x^{-2} x_2 \eta, \qquad H_2(\theta^*, Z) = x_1^2 - \sigma_x^2.$$

Because η has mean 0 and (x_t) and (η_t) are i.i.d. across time, all lead-lag covariances in $R(\theta^*)$ vanish and $R(\theta^*) = \mathrm{var}(H(\theta^*, Z))$ is diagonal with

$$R_{11}(\theta^*) = \mathrm{var}(\sigma_x^{-2} x_2 \eta) = \sigma_x^{-4}\,\mathbb{E}[x_2^2]\mathbb{E}[\eta^2] = \frac{\sigma_d^2}{\sigma_x^2}, \qquad R_{22}(\theta^*) = \mathrm{var}(x_1^2) = \mathbb{E}[x_1^4] - \sigma_x^4 = 2\sigma_x^4,$$

and $R_{12}(\theta^*) = 0$. Since B is diagonal, the integral for C is explicit:

$$C_{11} = \int_0^\infty e^{2B_{11}s} R_{11}\, ds = \frac{R_{11}}{-2B_{11}} = \frac{\sigma_d^2}{\sigma_x^2} \cdot \frac{1-\kappa}{2}, \qquad C_{22} = \int_0^\infty e^{2B_{22}s} R_{22}\, ds = \frac{R_{22}}{-2B_{22}} = \sigma_x^4, \qquad C_{12} = 0.$$

Consequently, for small γ and large t,

$$\theta_t = \begin{pmatrix} \beta_t \\ M_t \end{pmatrix} \approx N\left(\begin{pmatrix} 0 \\ \sigma_x^2 \end{pmatrix}, \gamma \begin{pmatrix} \frac{1-\kappa}{2} \frac{\sigma_d^2}{\sigma_x^2} & 0 \\ 0 & \sigma_x^4 \end{pmatrix} \right),$$

so in particular $\beta_t \approx N(0,\ \gamma\, \frac{1-\kappa}{2} \frac{\sigma_d^2}{\sigma_x^2})$ and $M_t \approx N(\sigma_x^2,\ \gamma\, \sigma_x^4)$.

B.3 Proof of proposition 3

Under soft-thresholded constant-gain learning, agents first compute the unpenalized weighted least squares estimate β_t^{OLS} exactly as in Section 5, then apply the soft-thresholding operator $S(\cdot, \lambda)$ before forming expectations. The key observation is that the updating rule for β_t^{OLS} remains unchanged—only the mapping from estimates to forecasts is modified. Formally, the learning dynamics are

$$\beta_t^{\mathrm{OLS}} = \beta_{t-1}^{\mathrm{OLS}} + \gamma M_{t-1}^{-1} x_{t-2}(r_{t-1} - x_{t-2}\beta_{t-1}^{\mathrm{OLS}}),$$

$$M_t = M_{t-1} + \gamma(x_{t-1}^2 - M_{t-1}),$$

and agents form forecasts using $\beta_t^{\mathrm{LASSO}} = S(\beta_t^{\mathrm{OLS}}, \lambda)$, which enters the actual law of motion as

$$r_t = \eta_t + \log\Big(1 - \kappa e^{w \tanh(x_{t-1} S(\beta_{t-1}^{\mathrm{OLS}}, \lambda)/w)}\Big) - \log\Big(1 - \kappa e^{w \tanh(x_t S(\beta_t^{\mathrm{OLS}}, \lambda)/w)}\Big).$$

The crucial insight is that β_t^{OLS} evolves according to the unpenalized system studied in Proposition 2. The soft-thresholding operator $S(\cdot, \lambda)$ affects prices and returns through the ALM, but the statistical properties of β_t^{OLS} itself are determined solely by the unpenalized updating rule. To verify that Proposition 2 applies, we check that all assumptions from the constant-gain analysis remain satisfied. The state-augmented system is unchanged, while the mapping H now includes the soft-thresholding operator through the return function $r(\cdot)$, but this does not affect regularity. Therfore it is enough to check assumptions (A.2) and (A.3′) for smoothness of H which hold because the soft-thresholding operator $S(\beta, \lambda)$ is Lipschitz continuous in β (with Lipschitz constant 1), and composition with the smooth $\tanh(\cdot)$ function preserves Lipschitz continuity of $r(\beta, s; \cdot)$ in (β, s). Since $\kappa e^w < 1$ by construction, the logarithmic terms remain uniformly bounded away from singularity. Thus H remains C^1 in θ with polynomial growth in Z.

The fixed point equations for the mean dynamics are now

$$\beta^{OLS*} = T(\beta^{L*}), \qquad \beta^{L*} = S(\beta^{OLS*}, \lambda).$$

In particular, $\beta^{OLS*} = 0$ implies $\beta^{L*} = 0$, so $(\beta^{OLS*}, M^*) = (0, \sigma_x^2)$ is always a fixed point. Any nonzero fixed point would require $|\beta^{OLS*}| > \lambda/\sigma_x^2$ and $\beta^{L*} = \beta^{OLS*} - \text{sign}(\beta^{OLS*})\lambda$, implying

$$\beta^{OLS*} = -\frac{\kappa}{1-\kappa}\Big(\beta^{OLS*} - \text{sign}(\beta^{OLS*})\lambda\Big) \;\Rightarrow\; \beta^{O*} = \text{sign}(\beta^{O*})\,\frac{\kappa}{1+\kappa}\,\lambda,$$

which contradicts $|\beta^{O*}| > \lambda$ for all $\kappa \in (0,1)$. Hence for $\kappa \in (0,1)$ the unique fixed point is $(0, \sigma_x^2)$ and the corresponding LASSO coefficient is $\beta^{L*} = 0$. Local stability follows from the same argument as above, since $\beta^L = S(\beta^O, \lambda/M)$ equals 0 in a neighborhood of $\beta^{OLS} = 0$. Since $\beta_t^L = S(\beta_t^{OLS}, \lambda)$ is a deterministic transformation, the implied approximate stationary law of the LASSO coefficient is the *soft–thresholded Gaussian* and

$$\mathbb{P}(|\beta_{\text{OLS}}| > \lambda) = 1 - \left[\Phi\left(\frac{\lambda}{\sigma_{\text{OLS}}}\right) - \Phi\left(\frac{-\lambda}{\sigma_{\text{OLS}}}\right)\right],$$

where $\sigma_{\text{OLS}} = \left(\gamma s_d^2\left(2\sigma_x^2\left(1+\frac{\kappa}{1-\kappa}\right)\right)^{-1}\right)^{1/2}$.

As a finite-sample benchmark for decreasing-gain learning, consider instead a fixed rolling window of length L, so each observation receives weight $1/L$. Under the null around CMCE, assume within a window that $r_i = \eta_i$ with $\eta_i \sim \mathcal{N}(0, \sigma_d^2)$, $x_i \sim \mathcal{N}(0, \sigma_x^2)$, and $x_i \perp \eta_i$. The OLS estimator is

$$\hat{\beta}_{t,L} = \frac{\sum_{i=1}^L x_i r_i}{\sum_{i=1}^L x_i^2} = \frac{\sigma_d}{\sigma_x\sqrt{L}}\,T_L,$$

where T_L is Student-t with L degrees of freedom. Therefore the probability of selecting a non-zero coefficient under soft-thresholding is

$$\mathbb{P}\Big(|\hat{\beta}_{t,L}| > \lambda\Big) = 2\left[1 - F_{t,L}\left(\frac{\lambda\,\sigma_x\sqrt{L}}{\sigma_d}\right)\right],$$

with $F_{t,L}$ the cdf of Student-t_L. For large L, this is approximately

$$\mathbb{P}\Big(|\hat{\beta}_{t,L}| > \lambda\Big) \approx 2\left[1 - \Phi\left(\frac{\lambda\,\sigma_x\sqrt{L}}{\sigma_d}\right)\right],$$

which decreases in both L and λ and converges to zero as $L \to \infty$.

C Additional Tables and Figures

Table 4: News Topics from Bybee et al. (2024)

Topic	Topic	Topic
Natural disasters	Internet	Soft drinks
Mobile devices	Profits	M&A
Changes	Police / crime	Research
Executive pay	Mid-size cities	Scenario analysis
Economic ideology	Middle East	Savings & loans
IPOs	Restraint	Electronics
Record high	Connecticut	Steel
Bond yields	Small business	Cable
Fast food	Disease	Activists
Competition	Music industry	Short sales
Nonperforming loans	Key role	News conference
U.S. defense	Political contributions	Revised estimate
Economic growth	Justice Department	Credit ratings
Broadcasting	Problems	Announce plan
Federal Reserve	Job cuts	Chemicals / paper
Regulation	Environment	Small caps
Unions	C-suite	Control stakes
Mutual funds	Venture capital	European sovereign debt
Mining	Company spokesperson	Private / public sector
Pharma	Schools	Russia
Programs / initiatives	Health insurance	Drexel
Trade agreements	Treasury bonds	Challenges
People familiar	Sales call	Publishing
Financial crisis	Aerospace / defense	Recession
Latin America	Cultural life	SEC
Earnings losses	Phone companies	Computers
Marketing	Japan	Nuclear / North Korea
NY politics	Tobacco	Product prices
Biology / chemistry / physics	Movie industry	Automotive
Machinery	Bankruptcy	Arts
International exchanges	Accounting	Space program
Immigration	Small changes	Small possibility
Agreement reached	Oil drilling	Rail / trucking / shipping
Indictments	Positive sentiment	Canada / South Africa
Airlines	California	Corporate governance
China	Investment banking	Spring / summer
Software	Pensions	Humor / language
Systems	Clintons	Major concerns
Mid-level executives	U.S. Senate	Agriculture
Bank loans	Takeovers	State politics
Real estate	Futures / indices	Southeast Asia
Optimism	Corrections / amplifications	Government budgets
Exchanges / composites	Currencies / metals	Mortgages
Financial reports	Germany	Rental properties

Continued on next page

Table 4 – *Continued from previous page*

Topic	**Topic**	**Topic**
Committees	Subsidiaries	Management changes
Share payouts	France / Italy	Acquired investment banks
Credit cards	Bear / bull market	Earnings forecasts
Terrorism	Watchdogs	Oil market
Couriers	Commodities	Utilities
Foods / consumer goods	Convertible / preferred	Macroeconomic data
Courts	Safety administrations	Reagan
Bush / Obama / Trump	Fees	Gender issues
Trading activity	Microchips	Insurance
Earnings	Luxury / beverages	Iraq
National security	Buffett	Taxes
Options / VIX	Casinos	Elections
Private equity / hedge funds	Negotiations	European politics
Size	NASD	Mexico
Retail	Long / short term	Wide range
Lawsuits	UK	Revenue growth

Table 5: Cross-Lag Correlations of Predictors

Lag k	Lag l	Mean corr.	Median corr.	Max |corr.|	P95 |corr.|
1	1	−0.0003	−0.0037	0.5453	0.0904
1	3	0.0003	−0.0001	0.2147	0.0567
1	5	0.0002	−0.0002	0.1653	0.0572
1	7	0.0001	−0.0004	0.1592	0.0567
1	9	0.0002	0.0000	0.1514	0.0552
1	11	0.0004	0.0001	0.2656	0.0598
3	1	0.0003	−0.0001	0.2147	0.0567
3	3	−0.0003	−0.0037	0.5456	0.0904
3	5	0.0003	−0.0001	0.2146	0.0568
3	7	0.0002	−0.0001	0.1660	0.0573
3	9	0.0001	−0.0004	0.1594	0.0568
3	11	0.0002	−0.0001	0.1513	0.0552
5	1	0.0002	−0.0002	0.1653	0.0572
5	3	0.0003	−0.0001	0.2146	0.0568
5	5	−0.0003	−0.0037	0.5455	0.0903
5	7	0.0003	−0.0000	0.2116	0.0568
5	9	0.0002	−0.0001	0.1658	0.0572
5	11	0.0001	−0.0004	0.1603	0.0566
7	1	0.0001	−0.0004	0.1592	0.0567
7	3	0.0002	−0.0001	0.1660	0.0573
7	5	0.0003	−0.0000	0.2116	0.0568
7	7	−0.0002	−0.0038	0.5456	0.0902
7	9	0.0003	−0.0000	0.2115	0.0568
7	11	0.0002	−0.0002	0.1649	0.0572
9	1	0.0002	0.0000	0.1514	0.0552
9	3	0.0001	−0.0004	0.1594	0.0568
9	5	0.0002	−0.0001	0.1658	0.0572
9	7	0.0003	−0.0000	0.2115	0.0568
9	9	−0.0002	−0.0037	0.5454	0.0902
9	11	0.0003	−0.0000	0.2130	0.0567
11	1	0.0004	0.0001	0.2656	0.0598
11	3	0.0002	−0.0001	0.1513	0.0552
11	5	0.0001	−0.0004	0.1603	0.0566
11	7	0.0002	−0.0002	0.1649	0.0572
11	9	0.0003	−0.0000	0.2130	0.0567
11	11	−0.0002	−0.0036	0.5453	0.0904

Notes: The table reports pairwise correlations between predictors at lags k and l. For each lag pair, we report the mean and median correlation across predictors, the maximum absolute correlation, and the 95th percentile of the absolute correlation distribution. Diagonal entries ($k = l$) reflect same-lag correlations, while off-diagonal entries capture cross-lag dependence.

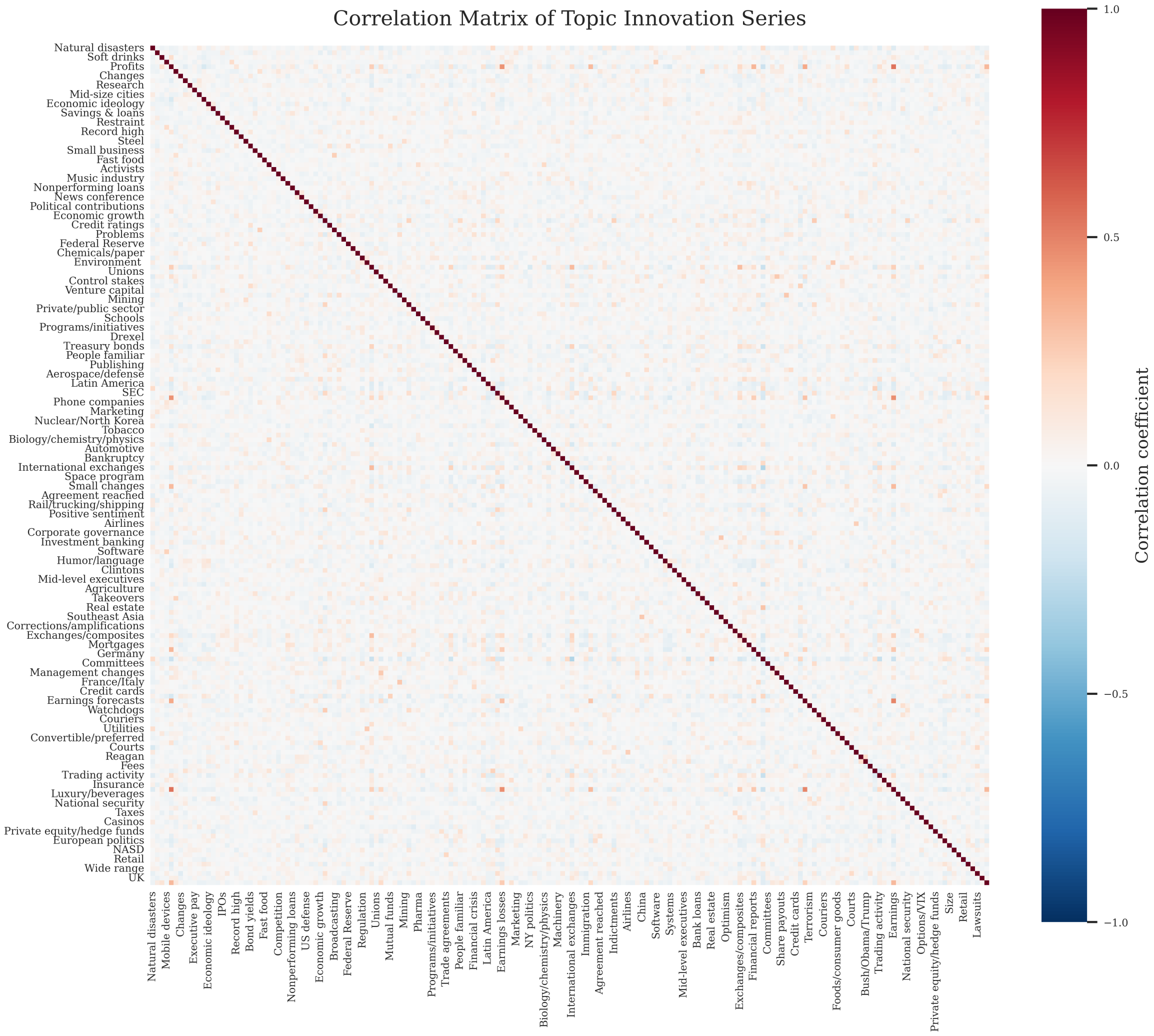

Figure 3: Correlation matrix of news-topic attention series.